

Spice Analyzer Report: Alfio_RAFFC_Pin_3

spice-analyzer

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1 Circuit under test

- Netlist: `tools/AnalogGym/.../spice_netlist/Alfio_RAFFC_Pin_3`
- Sizing: `design_variables/Alfio_RAFFC_Pin_3` (stock AnalogGym placeholders: $W=L=1\text{ }\mu\text{m}$ scale units)
- Class: three-stage OTA with **RAFFC** (reversed active-feedback frequency compensation)
- PDK / corner: `sky130_fd_pr 1.8 V nfet/pfet, tt`
- Supplies / bias: $V_{DD}=1.8\text{ V}$, $I_{\text{bias}}=20\text{ }\mu\text{A}$, $V_{\text{CM}}=300\text{ mV}$ (design), $C_L=100\text{ pF}$
- Compensation: $C_0=11\text{ pF}$ (outer, `net063→vout`), $C_1=0.35\text{ pF}$ (inner, `net050→net013`)

Topology. PMOS input pair (`xm8/xm9`, g_{m1}) with folded-cascode NMOS (`xm15/xm16` as g_{mb} / current buffer) and PMOS mirror load (`xm5/xm6`) forms stage 1 (output `net050`). Stage 2 is PMOS CS `xm10` (g_{m2}) loaded by NMOS current source `xm59` (output `net013`). Stage 3 is PMOS CS `xm7` (g_{m3}) with NMOS mirror `xm21/xm22` to `vout`. Feedforward PMOS `xm11` (g_{mf}) from `net050` to `vout`. Outer compensation C_0 closes through the cascode buffer (RAFFC active outer loop); C_1 is the reversed nested inner Miller.

2 Small-signal analysis

Assumptions: strong-inversion square-law estimates, $\mu_p C_{\text{ox}} \approx 50\text{ }\mu\text{A/V}^2$, $\mu_n C_{\text{ox}} \approx 270\text{ }\mu\text{A/V}^2$, $V_{\text{ov}} \approx 0.2\text{ V}$ when used, parasitics neglected, devices in saturation (idealized—see sim).

Bias. Reference $I_0=20\text{ }\mu\text{A}$. Tail `xm4` has $M=2 \Rightarrow I_{\text{tail}} \approx 40\text{ }\mu\text{A}$, so $I_{D1}=20\text{ }\mu\text{A}$ per input device.

$$g_{m1} \approx \sqrt{2\mu_p C_{\text{ox}}(W/L)I_{D1}} = \sqrt{2 \cdot 50 \times 10^{-6} \cdot 1 \cdot 20 \times 10^{-6}} \approx 45\text{ }\mu\text{S}$$

(alt. $g_{m1} \approx 2I_D/V_{\text{ov}} \approx 200\text{ }\mu\text{S}$).

Gains / GBW (RAFFC). With outer capacitor $C_c=C_0$:

$$\text{GBW} \approx \frac{g_{m1}}{2\pi C_0} \approx \begin{cases} 0.65\text{ MHz}, & g_{m1}=45\text{ }\mu\text{S} \\ 2.9\text{ MHz}, & g_{m1}=200\text{ }\mu\text{S}. \end{cases}$$

Three-stage DC gain (order-of-magnitude, $g_m r_o \sim 20\text{--}40$ dB per stage if saturated):

$$A_0 \approx A_1 A_2 A_3 \sim 60\text{ dB--}100\text{ dB}.$$

Phase margin: RAFFC places a LHP zero / buffered outer loop; with $C_1 \ll C_0$ and adequate $g_{m2,3}$, target PM is typically $\sim 60^\circ$ when sized—not computable reliably without valid OP g_m/r_o .

Power (order of magnitude). Branches at $\times 1\text{--}\times 8$ mirrors of $20\text{ }\mu\text{A}$ suggest $I_{DD} \sim 0.2\text{ mA--}1\text{ mA}$ $\Rightarrow P \sim 0.4\text{ mW--}1.8\text{ mW}$ at 1.8 V .

3 Simulation setup

- Deck: `results/Alfio_RAFFC_Pin_3.20260907_1745/tb/TB_Amplifier_ACDC.cir`
- Simulator: ngspice-47 (batch), sky130 tt 01v8 devices
- Analyses: DC temp $-40\text{--}125^\circ\text{C}$ (step 5°C), AC dec 20 from 0.1 Hz to 1 GHz
- Netlist snapshot: instance `m` folded into `w/mult` for ngspice/sky130
- Extra open-loop OP diagnostic: `TB_diag_OP.cir` at $V_{CM}=0.9\text{ V}$

4 Results

4.1 Operating point (open-loop diagnostic)

At $V_{CM}=0.9\text{ V}$: $V_{out} \approx 29\text{ mV}$ (railed low), $V(\text{net31}) \approx 1.799\text{ V}$ (PMOS tail $\approx$ shorted to V_{DD}), $V(\text{net013}) \approx 11\text{ mV}$, $V(\text{net050}) \approx 0.49\text{ V}$. Stock $W=L=1$ sizing does **not** establish a valid multi-stage saturation bias.

4.2 ADM AC / DC measures

Metric	Hand / small-signal	Simulated	Rel. error / notes
DC gain (dB)	$\sim 60\text{--}100$	-15.3	fail (not an amp OP)
GBW	$0.65\text{--}2.9\text{ MHz}$	(no 0 dB cross)	fail
Phase margin (deg)	~ 60 (design intent)	n/a	fail
Power	$0.4\text{--}1.8\text{ mW}$	0.326 mW	within rough band*
V_{os} @ 25°C	~ 0 (ideal)	-214 mV	fail

*Power match is coincidental at a collapsed bias point, not validation of the RAFFC design.

Other AC meas.: CM gain proxy `cmrrdc` = -29.1 dB ; `DCPSRp` = -14.9 dB ; `DCPSRn` = -3.0 dB .

ADM Bode. Magnitude stays near -15 dB from 0.1 Hz through midband, then falls to $\sim -100\text{ dB}$ by 1 GHz (data: `figures/bode_mag.dat`); DC phase $\approx 180^\circ$. No unity-gain frequency.

5 Match assessment

Match gates (DC gain ± 3 dB, GBW $\pm 30\%$, PM $\pm 10^\circ$): **all fail**.

Top causes:

1. **Placeholder sizing.** AnalogGym variables set every $W=L=1$ (and modest M). This is an optimization seed, not a biased RAFFC design; stage currents / headroom collapse (see OP).
2. **No unity-gain crossing.** With $|A_0| \approx -15$ dB, GBW and PM measures are undefined.
3. **Hand analysis assumes saturation** that the stock netlist does not achieve; theory numbers are topology targets, not predictions of this OP.

Recommendation. Re-size (or load a sized solution) so that g_{m1} – g_{m3} and cascode mirrors sit in saturation at $V_{CM} \approx 0.3$ – 0.9 V, then re-run this flow.

6 Appendix

Run id: Alfio_RAFFC_Pin_3_20260907_1745. Artifacts: `tb/`, `sim/ngspice.log`, `sim/diag.log`, `figures/bode_*.dat`.